

Home Module 4.py

localhost:8888/notebooks/Module%204.py

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JupyterLab Python 3 (ipykernel)

```
[2]:  
  
# Banking System while Loop- Ask for a pin until the correct pin is entered!  
correct_pin = "1234"  
entered_pin = input('Enter Password:')  
while entered_pin != correct_pin:  
    print('Incorrect pin! Please try again')  
    entered_pin = input('Enter Password:')  
print('Password accepted! Welcome:')
```

Enter Password: 111
Incorrect pin! Please try again
Enter Password: 1234
Password accepted! Welcome:

```
[3]:  
  
# VALIDATE USER INPUT: quantity (input()) while Q<=0 print quantity cannot not be zero or -ve  
# give user another chance else  
# print quantity saved display quantity  
quantity = int(input('Enter quantity:'))  
while quantity <=0:  
    print('Quantity cannot be zero or negative:')  
    quantity = int(input('Please enter a valid quantity:'))  
    print('Quantity saved:', quantity)
```

Enter quantity: -3
Quantity cannot be zero or negative:
Please enter a valid quantity: 0
Quantity saved: 0
Quantity cannot be zero or negative:
Please enter a valid quantity: 5
Quantity saved: 5

[4]:

```
# while loop to retard stock until its zero
stock = 5
while stock > 0:
    print('Items sold Remaining stock:', stock)
    stock -= 1
    print('Stock is finished. Reorder needed!')
```

```
Items sold Remaining stock: 5
Stock is finished. Reorder needed!
Items sold Remaining stock: 4
Stock is finished. Reorder needed!
Items sold Remaining stock: 3
Stock is finished. Reorder needed!
Items sold Remaining stock: 2
Stock is finished. Reorder needed!
Items sold Remaining stock: 1
Stock is finished. Reorder needed!
```

■ r 1.

[5]:

```
# Ride hailing app
driver_found = False
while not driver_found:
    print('Searching for a driver...')
    user_input = input('Driver found? (yes/no)')
    if user_input == 'yes':
        driver_found = True
print('Driver is on the way')
```

```
Searching for a driver...
Driver found? (yes/no) yes
Driver is on the way
```

[8]:

```
List = ['cement', 'bricks', 'tiles']
# Accessing Elements (index)
print(List[0]) # printing the first element of the list
print(List[-1])
```

```
cement
tiles
```

[10]:

```
# Slicing: Get the three trans for audit
transactions = [1500, -200, -350, 800, 120]
print(transactions[1:4])
# reverse list elements
print(transactions[::-1])
```

```
[-200, -350, 800]
[120, 800, -350, -200, 1500]
```

[11]:

```
products = ['cement', 'brick']
# add a new product
products.append(' Tiles')
for i in products:
    print(i)
```

```
cement
brick
 Tiles
```

```
#insert at a specific place
products.insert(0, 'paint')
print(products)
# remove an item
products.remove('Tiles')
```

```
['paint', 'cement', 'brick', ' Tiles']
```

[20]:

```
#insert at a specific place
products.insert(0, 'paint')
# remove an item
print(products)
```

```
['paint', 'paint', 'paint', 'paint', 'paint', 'cement', 'brick', ' Tiles']
```

[21]:

```
trans = [3500, -500, -200, 800]
trans.sort()
print('sorted', trans)
```

sorted [-500, -200, 800, 3500]

[24]:

```
# count number of entries
length = len(trans)
# sum
print(length)
```

4

•[25]:

```
# syntax of a function
def greet (name,):
    print(f'Hello, {name}')
greet('Fatima')
```

Hello, Fatima

[26]:

```
# syntax of a function
def greet (name,):
    print(f'Hello, {name} Your age is')
greet('Fatima')
```

Hello, Fatima Your age is

[11]:

```
# Function to cal uber fare
#distance * price per km R10
def uber_fare(distance, price_per_km):
    return distance*price_per_km
fare = uber_fare(40,10)
print(f'The total fare to be paid is {fare}')
```

The total fare to be paid is 400

[12]:

```
# Write a function that calculate Bank Account Balance
# 2 parameter (deposit withdrawal)
def calculate_balance(deposit, withdrawal):
    balance = deposit - withdrawal
    return balance
# Example usage
deposit = float(input('Enter deposit amount:'))
withdrawal = float(input('Enter withdrawal amount:'))

balance = calculate_balance(deposit, withdrawal)

print('Remaining balance:', balance)
```

```
Enter deposit amount: 1000
Enter withdrawal amount: 300
Remaining balance: 700.0
```

```
: # Courier System delivery time distance / speed
name = input('Enter name:')
Distance = float(input('Enter Delivery Distance (km):'))
speed = float(input('Enter speed (km/h):'))
def delivery_time(Distance, speed):
    time = Distance/speed
    return time
def msg(name):
    message = f'Hello {name}, Your order is ready for collection'
    return message
time = delivery_time(Distance, speed)
if time <= 1:
    print(f'Delivery time is {time:.2f}hrs. Your package will arrive shortly')
elif time <= 3:
    print(f'Delivery time is {time:.2f}hrs. Your package is on the way')
else:
    print(f'Delivery time is {time:.}hrs. Please be patient delivery may take longer than')
print(msg(name))
```

```
Enter name: John
Enter Delivery Distance (km): 60
Enter speed (km/h): 80
Delivery time is 0.75hrs. Your package will arrive shortly
Hello John, Your order is ready for collection
```